PhosphoAtlas Agent Benchmark

Summary Tables

Peptide accuracy = case-insensitive (biological identity match)

Table 1. Overall performance comparison across models (naive prompt condition).

Model	Type	Entries	Recall	Prec	F1	Kinases	Pep Acc	Multi-DB	DBs Used
Claude Sonnet 4.6	Proprietary	15,434	0.8727	0.9007	0.8865	404/433	100.0%	0.0%	PhosphoSitePlus
claude_sonnet_paper_info	Unknown	17,715	0.8797	0.7937	0.8345	406/433	99.1%	8.3%	PhosphoSitePlus, UniProt
claude_sonnet_pipeline_g	Unknown	17,655	0.8780	0.7948	0.8343	404/433	99.2%	8.1%	PhosphoSitePlus, UniProt
Claude Opus 4.6	Proprietary	23,342	0.9529	0.6504	0.7731	418/433	99.7%	28.6%	PhosphoSitePlus, SIGNOR, UniPr
Qwen 3.2 35B	Open-source	9,671	0.4629	0.7483	0.5720	377/433	95.6%	0.0%	SIGNOR
GPT-5 (OpenCLAW)	Proprietary	11,142	0.4826	0.6832	0.5656	382/433	0.0%	0.0%	PhosphoSIGNOR
Gemini 3.1 Pro	Proprietary	1,428	0.0500	0.5476	0.0916	61/433	0.0%	8.3%	N/A
Mistral Large	Proprietary	15	0.0008	0.8000	0.0016	9/433	0.0%	33.3%	PhosphoSitePlus, SIGNOR
Gemini 2.5 Flash	Proprietary	9	0.0000	0.0000	0	0/433	0	0.0%	UniProt, UniProtKB
Gemini 3.0 Flash	Proprietary	9	0.0000	0.0000	0	0/433	0	0.0%	UniProt, UniProtKB

Table 2. Effect of prompt engineering on model performance.

Model	Condition	Recall	F1	Pep Acc	Multi-DB	vs Naive
Claude Opus 4.6	naive	0.9529	0.7731	99.7%	28.6%	--
Claude Opus 4.6	paper_informed	0.9529	0.7731	99.7%	28.6%	No change
Claude Opus 4.6	pipeline_guided	0.9529	0.7731	99.7%	28.6%	No change
Gemini 3.1 Pro	naive	0.0500	0.0916	0.0%	8.3%	--
Gemini 3.1 Pro	paper_informed	0.0404	0.0760	0.0%	0.0%	-0.010 R
Gemini 3.1 Pro	pipeline_guided	0.0309	0.0591	0.0%	6.3%	-0.019 R
Mistral Large	naive	0.0008	0.0016	0.0%	33.3%	--
Mistral Large	paper_informed	0.0003	0.0006	0.0%	0.0%	No change
Qwen 3.2 35B	explicit_prompt	0.4730	0.5720	85.5%	0.0%	+0.010 R
Qwen 3.2 35B	naive	0.4629	0.5720	95.6%	0.0%	--
Qwen 3.2 35B	paper_informed	0.4629	0.5720	95.6%	0.0%	No change
Qwen 3.2 35B	pipeline_informed	0.4730	0.5720	85.5%	0.0%	+0.010 R

Table 3. Claude Opus vs Sonnet adjusted comparison (assuming FPs are valid novel entries).

Metric	Opus	Sonnet	Winner
Gold entries found (TP)	14,899	13,644	Opus (+1,255)
Novel entries (likely valid)	8,010	1,505	Opus (+6,505)
Total real data curated	22,909	15,149	Opus (+7,760)
Kinases found	418/433	404/433	Opus
Multi-DB cross-validation	28.6%	0.0%	Opus
Peptide accuracy (case-insensitive)	99.7%	100.0%	Comparable
Peptide true mismatches	34/14106	7/13601	Sonnet
UniProt accuracy	91.9%	99.7%	Comparable

Table 4. Tool call efficiency across models and conditions.

Model	Condition	Tool Calls	Runtime	DBs Queried	Entries
Claude Opus 4.6	naive	24	45s	PhosphoSitePlus, SIGNOR, UniPr	23,342
Claude Opus 4.6	paper_informed	21	36s	PhosphoSitePlus, SIGNOR, UniPr	23,342
Claude Opus 4.6	pipeline_guided	21	35s	PhosphoSitePlus, SIGNOR, UniPr	23,342
Claude Sonnet 4.6	naive	31	10.0m	PhosphoSitePlus	15,434
claude_sonnet_paper_info	naive	?	6.3m	PhosphoSitePlus, UniProt	17,715
claude_sonnet_pipeline_g	naive	?	84s	PhosphoSitePlus, UniProt	17,655
Gemini 3.1 Pro	naive	?	?	N/A	1,428
Gemini 2.5 Flash	naive	?	?	UniProt, UniProtKB	9
Gemini 3.0 Flash	naive	?	?	UniProt, UniProtKB	9
Gemini 3.1 Pro	paper_informed	?	?	N/A	979
Gemini 3.1 Pro	pipeline_guided	?	?	N/A	718
Mistral Large	naive	3	110s	PhosphoSitePlus, SIGNOR	15
Mistral Large	paper_informed	4	91s	PhosphoSitePlus	5
GPT-5 (OpenCLAW)	naive	?	?	PhosphoSIGNOR	11,142

Table 5. Per-tier recall by kinase size (naive prompt condition).

Model	Tier A (100+)	Tier B (20-99)	Tier C (5-19)	Tier D (<5)
Claude Sonnet 4.6	0.8811	0.8626	0.8698	0.7700
claude_sonnet_paper_informed	0.8880	0.8693	0.8788	0.7764
claude_sonnet_pipeline_guided	0.8876	0.8691	0.8636	0.7764
Claude Opus 4.6	0.9618	0.9403	0.9456	0.8946
Qwen 3.2 35B	0.4626	0.4496	0.4924	0.5176
GPT-5 (OpenCLAW)	0.4848	0.4657	0.5048	0.5463
Gemini 3.1 Pro	0.0552	0.0404	0.0530	0.0128
Mistral Large	0.0013	--	--	--
Gemini 2.5 Flash	--	--	--	--
Gemini 3.0 Flash	--	--	--	--

Table 6. Column-level accuracy for matched entries (naive prompt condition). Pep Acc = case-insensitive match rate an

Model	Matched	Pep Acc*	Hallucinated	Missing	Pep Total	UniProt Acc
Claude Sonnet 4.6	13,644	99.9%	7	0	13601	99.7%
claude_sonnet_paper_info	13,754	99.9%	7	110	13641	99.7%
claude_sonnet_pipeline_g	13,727	99.9%	7	107	13615	99.7%
Claude Opus 4.6	14,899	99.8%	34	11	14106	91.9%
Qwen 3.2 35B	7,237	95.6%	283	0	6454	98.9%
GPT-5 (OpenCLAW)	7,545	0.0%	0	6760	6760	98.7%
Gemini 3.1 Pro	782	0.0%	521	254	775	96.2%
Mistral Large	12	0.0%	12	0	12	91.7%
Gemini 2.5 Flash	0	0.0%	0	0	0	0
Gemini 3.0 Flash	0	0.0%	0	0	0	0

Table 7. Token usage and computational cost.

Model	Condition	Input Tok	Output Tok	Total Tok	Tool Calls	Est. Cost
Claude Opus 4.6	naive	1,485,032	11,081	1,496,113	24	$23.11
Claude Opus 4.6	paper_informed	1,280,027	9,803	1,289,830	21	$19.94
Claude Opus 4.6	pipeline_guided	1,280,742	9,803	1,290,545	21	$19.95
Claude Sonnet 4.6	naive	258,072	13,428	271,500	31	$0.98
claude_sonnet_paper_info	naive	891,947	11,582	903,529	?	$2.85
claude_sonnet_pipeline_g	naive	1,021,544	11,345	1,032,889	?	$3.24